

Gauss's Central-Difference formulas

(a) Gauss's forward formula: In the general Newton

formula, put $x_0 = x_0$, $x_1 = x_0 + h$, $x_2 = x_0 - h$, $x_3 = x_0 + 2h$, $x_4 = x_0 - 2h$,
 $x_5 = x_0 + 3h$, $x_6 = x_0 - 3h$ etc.

Then we have,

$$y = y_0 + (x - x_0) \delta(x_0, x_1) + (x - x_0)(x - x_1) \delta(x_0, x_1, x_2) + \dots$$

$$= y_0 + (x - x_0) \delta(x_0, x_0 + h) + (x - x_0)(x - x_0 - h) \delta(x_0, x_0 + h, x_0 - h) + \\ + (x - x_0)(x - x_0 - h)(x - x_0 + h) \delta(x_0, x_0 + h, x_0 - h, x_0 + 2h, \\ (x - x_0)(x - x_0 - h)(x - x_0 + h) (x - x_0 - 2h) \delta(x_0, x_0 + h, x_0 - h, x_0 + 2h, \\ x_0 - 2h) + (x - x_0)(x - x_0 - h)(x - x_0 + h)(x - x_0 - 2h) \\ (x_0 - x_0 + 2h) \delta(x_0, x_0 + h, x_0 - h, x_0 + 2h, x_0 - 2h, x_0 + 3h) + \dots$$

$$x - x_0 = hu$$

$$\Rightarrow x = x_0 + hu$$

Now put $u = \frac{x - x_0}{h}$, on $x - x_0 = hu$ then

$$\begin{aligned} \Delta^3 y_0 = & y_0 + hu \cdot \delta(x_0, x_0+h) + hu(hu-h) \delta(x_0-h, x_0, x_0+h) + \\ & hu(hu-h)(hu+h) \delta(x_0-h, x_0, x_0+h, x_0+2h) + hu(hu-h) \\ & (hu+h)(hu-2h) \delta(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) + hu(hu-h) \\ & (hu+h)(hu-2h)(hu+2h) \delta(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h, x_0+3h) \end{aligned}$$

$$\text{--- (2)}$$

But we have, $\delta(x_0-h, x_0, x_0+h) = \delta(x_0+h, x_0, x_0-h)$

$$\delta(x_0, x_0+h) = \frac{\Delta y_0}{h}, \quad \delta(x_0-h, x_0, x_0+h) = \frac{\delta(x_0+h, x_0) - \delta(x_0, x_0-h)}{x_0+h - x_0-h}$$

$$= \frac{\frac{\Delta y_0}{h} - \frac{\Delta y_0}{h}}{2h} = \frac{\Delta^2 y_0}{2h^2}$$

$$\therefore \delta(x_0-h, x_0, x_0+h) = \frac{\Delta^2 y_{-1}}{2h^2}$$

$$\delta(x_0-h, x_0, x_0+h, x_0+2h) = \frac{\delta(x_0-h, x_0, x_0+h) - \delta(x_0, x_0+h, x_0+2h)}{x_0-h - x_0-2h}$$

$$= \frac{\delta(x_0, x_0+h, x_0+2h) - \delta(x_0-h, x_0, x_0+h)}{2h}$$

$$= \frac{\frac{\Delta^2 y_1}{2h^2} - \frac{\Delta^2 y_{-1}}{2h^2}}{2h} = \frac{\Delta^3 y_0}{2h^3}$$

$$s(x_0-h, x_0, x_0+h, x_0+2h) = \frac{\Delta^3 y_1 - \Delta^3 y_0}{6h^3}$$

$$s(x_0, x_0+h, x_0+2h) = \frac{s(x_0, x_0+h, x_0+2h) - s(x_0, x_0+h)}{2h} = \frac{s(x_0+h, x_0+2h) - s(x_0, x_0+h)}{2h}$$

$$= \frac{\frac{\Delta^2 y_1}{2h} - \frac{\Delta^2 y_0}{2h}}{2h} = \frac{\Delta^2 y_1 - \Delta^2 y_0}{2h^2}$$

$$= \frac{\frac{\Delta y_1}{h} - \frac{\Delta y_0}{h}}{2h} = \frac{\Delta y_1 - \Delta y_0}{2h^2}$$

$$\therefore s(x_0-h, x_0, x_0+h, x_0+2h) = \frac{\frac{\Delta y_1}{h} - \frac{\Delta y_0}{h}}{2h} = \frac{\Delta y_1 - \Delta y_0}{2h^2}$$

$$= \frac{\Delta^3 y_1 - \Delta^3 y_0}{6h^3}$$

$$= \frac{\Delta^3 y_1}{3! h^3}$$

Similarly,

$$s(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) = \frac{\Delta^4 y_2}{4! h^4}$$

$$s(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h, x_0+3h) = \frac{\Delta^5 y_2}{5! h^5}$$

substituting these values in (12), we get

$$y = y_0 + hu \frac{\Delta y_0}{h} + hu(hu-h) \frac{\Delta^2 y_{-1}}{2! h^2} + hu(hu-h)(hu+h) \frac{\Delta^3 y_{-1}}{3! h^3} +$$

$$hu(hu-h)(hu+h)(hu-2h) \frac{\Delta^4 y_{-2}}{4! h^4} + hu(hu-h)(hu+h)$$

$$(hu-2h)(hu+2h) \frac{\Delta^5 y_{-2}}{5! h^5} + \dots$$

$$= y_0 + u \Delta y_0 + u(u-1) \frac{\Delta^2 y_{-1}}{2!} + u(u-1)(u+1) \frac{\Delta^3 y_{-1}}{3!} + u(u-1)(u+1)(u-2)(u+2) \frac{\Delta^4 y_{-2}}{4!} + \dots$$

$$= y_0 + u \Delta y_0 + u(u-1) \frac{\Delta^2 y_{-1}}{2!} + u(u-1) \frac{\Delta^3 y_{-1}}{3!} + u(u-1)(u-2) \frac{\Delta^4 y_{-2}}{4!} + \dots$$

This is Gauss's forward formula.

(b) Gauss's backward formula: To derive the Gauss backward formula, make the following substitutions to the Newton's formula

$$y = y_0 + (x-x_0) \delta(x_0, x_0-h) + (x-x_0)(x-x_1) \delta(x_0, x_0-h, x_0+h) +$$

$$y = y_0 + (x-x_0) \delta(x_0, x_1) + (x-x_0)(x-x_1) \delta(x_0, x_1, x_2) + (x-x_0)(x-x_1)(x-x_2)$$

$$\delta(x_0, x_1, x_2, x_3) + \dots$$

$$x_0 = x_0, \quad x_1 = x_0 - h, \quad x_2 = x_0 + h, \quad x_3 = x_0 - 2h, \quad x_4 = x_0 + 2h, \quad x_5 = x_0 - 3h, \\ x_6 = x_0 + 3h \text{ etc.}$$

Then Newton formula becomes

$$y = y_0 + (x-x_0) \delta(x_0, x_0-h) + (x-x_0)(x-x_0+h) \delta(x_0, x_0-h, x_0+h) + \\ (x-x_0)(x-x_0+h)(x-x_0-h) \delta(x_0, x_0-h, x_0+h, x_0-2h) + \\ (x-x_0)(x-x_0+h)(x-x_0-h)(x-x_0+2h) \delta(x_0, x_0-h, x_0+h, x_0-2h, x_0+2h) \\ + (x-x_0)(x-x_0+h)(x-x_0-h)(x-x_0+2h)(x-x_0-2h) \delta(x_0, x_0-h, x_0+h, \\ x_0-2h, x_0+2h, x_0-3h) + \dots$$

$$= y_0 + (x-x_0) \delta(x_0-h, x_0)$$

$$\text{Now put } u = \frac{x-x_0}{h},$$

$$\Rightarrow x = x_0 + uh, \text{ then}$$

$$y = y_0 + (x_0 + uh - x_0) \delta(x_0-h, x_0) + (x_0 + uh - x_0)(x_0 + uh - x_0 + h) \\ \delta(x_0-h, x_0, x_0+h) + (x_0 + uh - x_0)(x_0 + uh - x_0+h)(x_0 + uh - x_0-h) \\ \delta(x_0-2h, x_0-h, x_0, x_0+h) + (x_0 + uh - x_0)(x_0 + uh - x_0+h) \\ (x_0 + uh - x_0-h)(x_0 + uh - x_0+2h) \delta(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) \\ + (x_0 + uh - x_0)(x_0 + uh - x_0+h)(x_0 + uh - x_0-h)(x_0 + uh - x_0+2h)(x_0 + uh - x_0-2h) \\ \delta(x_0-3h, x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) + \dots$$

$$\begin{aligned}
 y &= y_0 + uh \delta(x_0-h, x_0) + uh (uh+h) \delta(x_0-h, x_0, x_0+h) + uh (uh+h) \\
 &\quad (uh-h) \delta(x_0-2h, x_0-h, x_0, x_0+h) + uh (uh+h) (uh-h) \\
 &\quad (uh+2h) \delta(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) \\
 &\quad + uh (uh+h) (uh-h) (uh+2h) (uh-2h) \delta(x_0-3h, x_0-2h, x_0-h, x_0, x_0+h, x_0+2h, x_0+3h)
 \end{aligned}$$

But $\delta(x_0-h, x_0) = \frac{\Delta y_{-1}}{h}$, $\delta(x_0-h, x_0, x_0+h) = \frac{\delta(x_0-h, x_0) - \delta(x_0, x_0+h)}{x_0-h-x_0-h}$

$$= \frac{\frac{\Delta y_{-1}}{h} - \frac{\Delta y_0}{h}}{-2h}$$

$$= \frac{\Delta y_0 - \Delta y_{-1}}{2h^2}$$

$$\therefore \delta(x_0-h, x_0, x_0+h) = \frac{\Delta y_{-1}}{2h^2}$$

$$\delta(x_0-2h, x_0-h, x_0, x_0+h) = \frac{\Delta^2 y_{-2}}{3! h^3}$$

$$\delta(x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) = \frac{\Delta^3 y_{-2}}{4! h^4}$$

$$\delta(x_0-3h, x_0-2h, x_0-h, x_0, x_0+h, x_0+2h) = \frac{\Delta^4 y_{-3}}{5! h^5}$$

substituting these values in (2) and we get

$$y = y_0 + uh \frac{\Delta^2 y_{-1}}{2} + uh \cdot h(u+1) \frac{\Delta^3 y_{-1}}{24} + uh \cdot h(u+1) \cdot h(u+2) \frac{\Delta^4 y_{-2}}{4! h^4} + uh \cdot h(u+1) \cdot h(u-1) \frac{\Delta^3 y_{-2}}{3! h^3} +$$

$$h(u-2) \frac{\Delta^5 y_{-3}}{5! h^5} + \dots$$

$\Delta + 1 = 3$, fourth order

$$= y_0 + u \Delta y_{-1} + u(u+1) \frac{\Delta^2 y_{-1}}{2!} + u(u+1)(u-1) \frac{\Delta^3 y_{-2}}{3!} + u(u+1)(u-1)(u+2) \frac{\Delta^4 y_{-3}}{4!} +$$

This is Gauss's backward formula.